Chapter 1: Vectors

Tutorial

Exercise 1: Free Vectors

1. $\overrightarrow{PQ} + \overrightarrow{QR} + \overrightarrow{RS} + \overrightarrow{ST}$ PT
2. $\overrightarrow{AC} + \overrightarrow{CL} - \overrightarrow{ML}$ AM
3. $\overrightarrow{GH} + \overrightarrow{HJ} + \overrightarrow{JK} + \overrightarrow{KL} + \overrightarrow{LG}$ 0
4. $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} + \overrightarrow{DB}$ AB

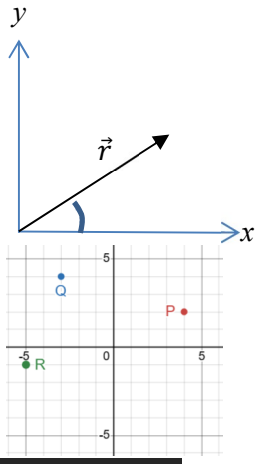
Exercise 2: Position Vectors

1. The x component of vector $\vec{A}$ is -25.0 m and the y component is $+40.0$ m.

   (a) What is the magnitude of $\vec{A}$? $\sqrt{((-25)^2 + (40)^2)} = \sqrt{2225}$

   (b) What is the angle between the direction of $\vec{A}$ and the positive direction of x? $180 - \tan^{-1}\left(\frac{40}{25}\right) = 122.01°$

2. A displacement vector $\vec{r}$ in the xy plane is 15 m long and directed at angle $\theta = 30°$ as shown in the figure. Determine the x- and y- component of the vector.



$x: 15cos30° = 7.5\sqrt{3}\,\hat{i}\; y: 15sin30° = 7.5\,\hat{j}$

3. Given the points P(4, 2), Q(–3, 4) and R(–5– ,1)

   a) Plot the points on graph paper, and draw and label position vectors $\overrightarrow{O\,P}, \overrightarrow{O\,Q}$ and $\overrightarrow{O\,R}$

   b) Write position vectors $\overrightarrow{O\,P}, \overrightarrow{O\,Q}$ and $\overrightarrow{O\,R}$ $\overrightarrow{OP} = 4\hat{i} + 2\hat{j}$, $\overrightarrow{OQ} = -3\hat{i} + 4\hat{j}$, $\overrightarrow{OR} = -5\hat{i} - \hat{j}$

   c) Find the vectors $\overrightarrow{Q\,P}, \overrightarrow{R\,Q}$ and $\overrightarrow{P\,R}$ $\overrightarrow{QP} = -\overrightarrow{OQ} + \overrightarrow{OP} = 4\hat{i} + 2\hat{j} - (-3\hat{i} + 4\hat{j}) = 7\hat{i} - 2\hat{j}$, $\overrightarrow{RQ} = \overrightarrow{OQ} - \overrightarrow{OR} = -3\hat{i} + 4\hat{j} - (-5\hat{i} - \hat{j}) = 2\hat{i} + 5\hat{j}$, $\overrightarrow{PR} = -\overrightarrow{OP} + \overrightarrow{OR} = -5\hat{i} - \hat{j} - (4\hat{i} + 2\hat{j}) = -9\hat{i} - 3\hat{j}$

   d) Find (i) $|\overrightarrow{Q\,P}|$ $r = \sqrt{7^2 + -2^2} = \sqrt{53}$

   (ii) $|\overrightarrow{R\,Q}|$ $r = \sqrt{2^2 + 5^2} = \sqrt{29}$

   (iii) $|\overrightarrow{P\,R}|$ $r = \sqrt{-9^2 + -3^2} = \sqrt{90}$

4. Let $z_1 = 2\hat{i} + 4\hat{j}$ and $z_2 = 5\hat{i} + 2\hat{j}$. Find $z_1 + z_2$ and $z_1 - z_2$. $z_1 + z_2 = 2\hat{i} + 4\hat{j} + 5\hat{i} + 2\hat{j} = 7\hat{i} + 6\hat{j}$, $z_1 - z_2 = 2\hat{i} + 4\hat{j} - (5\hat{i} + 2\hat{j}) = -3\hat{i} + 2\hat{j}$
5. If $z_1 = 5\hat{i} - 2\hat{j}$, $z_2 = 3\hat{i} + 3\hat{j}$ and $z_3 = 4\hat{i} - \hat{j}$. Find $z_1 + z_2 + z_3$ and $z_1 - z_2 - z_3$. $z_1 + z_2 + z_3 = 5\hat{i} - 2\hat{j} + 3\hat{i} + 3\hat{j} + 4\hat{i} - \hat{j} = 12\hat{i}$, $z_1 - z_2 - z_3 = 5\hat{i} - 2\hat{j} - (3\hat{i} + 3\hat{j}) - (4\hat{i} - \hat{j}) = -2\hat{i} - 4\hat{j}$
6. If $\overrightarrow{O\,A} = 3\hat{i} + 5\hat{j}$ and $\overrightarrow{O\,B} = 5\hat{i} - 2\hat{j}$, find $\overrightarrow{A\,B}$ $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} = 5\hat{i} - 2\hat{j} - (3\hat{i} + 5\hat{j}) = 2\hat{i} - 7\hat{j}$
7. Express $\overrightarrow{A\,B}$ in the form $x\hat{i} + y\hat{j} + z\hat{k}$ for:
   (a) A(0, 3, 2) and B(–1, – 4, 3)
   (b) A(1, 4, 1) and B(1, 4, 6)

8. Find the values of x and y if:
   (a) $\underset{\sim}{a} = 4\hat{\imath} - \hat{\jmath}, \quad \underset{\sim}{b} = x\hat{\imath} + 3y\hat{\jmath}, \quad \underset{\sim}{a} + \underset{\sim}{b} = 7\hat{\imath} - 2\hat{\jmath}$
   (b) $\underset{\sim}{a} = x\hat{\imath} + 3\hat{\jmath}, \quad \underset{\sim}{b} = -2\hat{\imath} + 5y\hat{\jmath}, \quad \underset{\sim}{a} - \underset{\sim}{b} = 6\hat{\imath} + \hat{\jmath}$

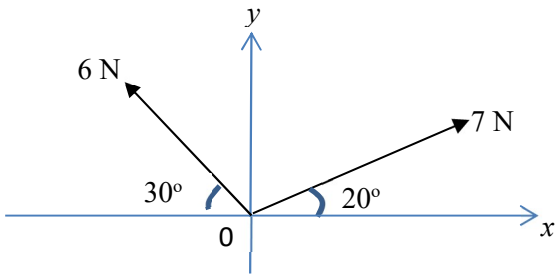
9. A, B, C and D are points defined respectively by the position vectors, $\underset{\sim}{a} = \hat{\imath} + 3\hat{\jmath} - 2\hat{k}, \quad \underset{\sim}{b} = 5\hat{\imath} + \hat{\jmath} - 6\hat{k}, \quad \underset{\sim}{c} = 5\hat{\jmath} + 3\hat{k}$ and $\underset{\sim}{d} = 2\hat{\imath} + 4\hat{\jmath} + \hat{k}$
   (a) Find : (i) $\overrightarrow{AB}$
   (ii) $\overrightarrow{BC}$
   (iii) $\overrightarrow{CD}$
   (iv) $\overrightarrow{DA}$
   (b) Evaluate: (i) $\left|\overrightarrow{AC}\right|$
   (ii) $\left|\overrightarrow{BD}\right|$

10. Given $A(2,1)$, $B(1,-3)$, $C(-5,2)$ and $D(3,5)$ are coordinates in an x-y plane and O is the origin.
    1. Find: (i) $\overrightarrow{OA}$
    (ii) $\overrightarrow{AB}$
    (iii) $\overrightarrow{BC}$
    (iv) $\overrightarrow{BD}$
    (b) Show that $\overrightarrow{AB}$ and $\overrightarrow{BD}$ are parallel.
    (c) What can be said of the points A, B and D?

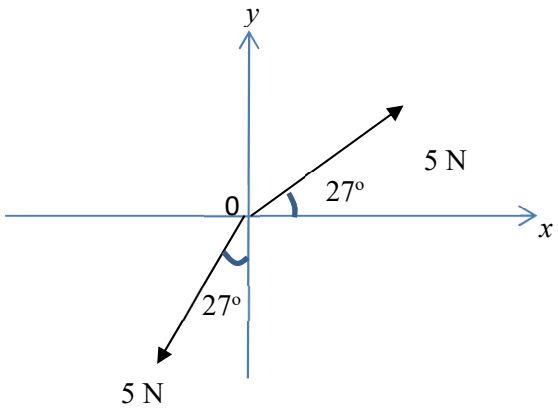
11. The forces $\overrightarrow{F_1} = (3\hat{\imath} + 2\hat{\jmath})N$, $\overrightarrow{F_2} = (6\hat{\imath} - 4\hat{\jmath})N$ and $\overrightarrow{F_3} = (2\hat{\imath} - \hat{\jmath})N$ act on a particle. Find the resultant force action on the particle.

12. For each of the following diagram, find the resultant force expressed (i) in unit vector notation, (ii) as magnitude and direction. (The angle is measured anticlockwise from the positive x axis)

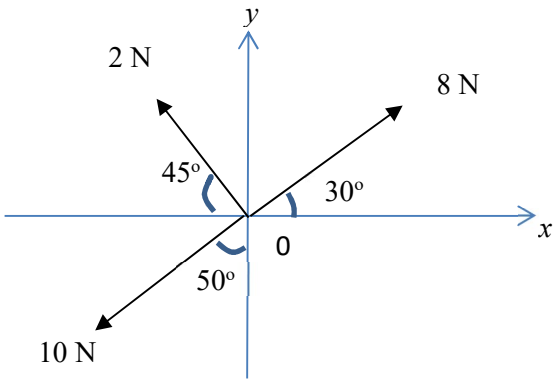
(a)



(b)



(c)



Exercise 3: Unit vectors in the direction of a given vector and Direction Cosines

1. For vectors $\underset{\sim}{\mathbf{a}}$ and $\underset{\sim}{\mathbf{b}}$ where $\underset{\sim}{\mathbf{a}} = -2\hat{\imath} + 3\hat{\jmath}$ and $\underset{\sim}{\mathbf{b}} = 4\hat{\imath} - 5\hat{\jmath}$ find :
   - (a) $|\underset{\sim}{\boldsymbol{b}}|$
   - (b) $\hat{\underset{\sim}{\mathbf{b}}}$
   - (c) The angle $\underset{\sim}{\mathbf{b}}$
   - (d) The position vector of point A relative to point B, i.e. $\overrightarrow{BA}$

2. Given the points P (3, –1, 4), Q (2, 3,1) and R (1, –3, –2) find:
   - (a) $\underset{\sim}{\mathbf{p}}$, $\underset{\sim}{\mathbf{q}}$ and $\underset{\sim}{\mathbf{r}}$ (the position vectors of P, Q and R)
   - (b) $\hat{\underset{\sim}{\mathbf{q}}}$
   - (c) $\underset{\sim}{\mathbf{p}} + \underset{\sim}{\mathbf{q}}$
   - (d) $2\underset{\sim}{\mathbf{r}} - \underset{\sim}{\boldsymbol{p}}$
   - (e) the direction cosines for $\underset{\sim}{\mathbf{r}}$
   - (f) the angle which $\underset{\sim}{\mathbf{p}}$ makes with each of the coordinate axes.
   - (g) $\overrightarrow{\boldsymbol{PQ}}$
   - (h) $\overrightarrow{\boldsymbol{RP}}$
   - (i) $|2\underset{\sim}{\boldsymbol{p}}+3\underset{\sim}{\mathbf{r}}|$

3. Find the unit vector which has the same direction as $\overrightarrow{\boldsymbol{OA}}$ where A is (5, 2, –1).
4. If $\underset{\sim}{p} = 2\hat{\imath} - 3\hat{\jmath} + \hat{k}$, $\underset{\sim}{q} = 2\hat{\imath} - 4\hat{\jmath} + 5\hat{k}$, $\underset{\sim}{r} = -\hat{\imath} - 4\hat{\jmath} + 2\hat{k}$, find the values of scalar constants m and n such that $\underset{\sim}{p} + m\underset{\sim}{q} + n\underset{\sim}{r}$ is parallel to the x-axis.
5. Let $\underset{\sim}{a} = 5\hat{\imath} + 2\hat{\jmath}$, $\underset{\sim}{b} = 2\hat{\imath} - 3\hat{\jmath}$, $\underset{\sim}{c} = 2\hat{\imath} + \hat{\jmath} + \hat{k}$ and $\underset{\sim}{d} = -\hat{\imath} + 4\hat{\jmath} + 2\hat{k}$

   (a) Find:
   - (i) $|\underset{\sim}{a}|$
   - (ii) $|\underset{\sim}{b}|$
   - (iii) $|\underset{\sim}{a} + 2\underset{\sim}{b}|$
   - (iv) $|\underset{\sim}{c} - \underset{\sim}{d}|$

   (b) Find correct to 2 decimal places the angle which each of the following vectors makes with the positive direction of the x – axis :
   - (i) $\underset{\sim}{a}$
   - (ii) $\underset{\sim}{a} + 2\underset{\sim}{b}$
   - (iii) $\underset{\sim}{c} - \underset{\sim}{d}$

Exercise 4: Vector Operation: Dot Product

1. If $\underset{\sim}{p} = 2\,\hat{\imath} - 3\,\hat{\jmath} + \hat{k}$, $\underset{\sim}{q} = 2\,\hat{\imath} + 4\,\hat{\jmath} - 5\,\hat{k}$, $\underset{\sim}{r} = -\hat{\imath} - 2\,\hat{\jmath} - 4\,\hat{k}$ find

   (a) $\underset{\sim}{\mathbf{p}} \cdot \underset{\sim}{\boldsymbol{q}}$

   (b) $\underset{\sim}{\mathbf{q}} \cdot \underset{\sim}{\boldsymbol{p}}$

   (c) the angle between $\underset{\sim}{\mathbf{p}}$ and $\underset{\sim}{\mathbf{q}}$

   (d) prove that $\underset{\sim}{\mathbf{p}}$ and $\underset{\sim}{\mathbf{r}}$ are perpendicular.

   (e) $\left|\underset{\sim}{\boldsymbol{p}}\right|$

   (f) $\underset{\sim}{\mathbf{p}} \cdot \underset{\sim}{\boldsymbol{p}}$

   (g) $\underset{\sim}{\mathbf{p}} \cdot \left(\underset{\sim}{\boldsymbol{r}} - \underset{\sim}{\boldsymbol{q}}\right)$

2. If $\underset{\sim}{q} = 3\hat{\imath} + 2\hat{\jmath}$ and $\underset{\sim}{r} = c\hat{\imath} - 2\,\hat{\jmath}$ find the value of c when $\underset{\sim}{\boldsymbol{q}}$ and $\underset{\sim}{\mathbf{r}}$ are at right angles to each other.
3. Given vectors $\underset{\sim}{\boldsymbol{a}} = 4\,\hat{\imath} - 5\hat{\jmath}$ and $\underset{\sim}{\mathbf{b}} = \hat{\imath} + 3\,\hat{\jmath}$ find the angle between the vectors.
4. When constant force of 35 newton acts on an object, it does 385 joule of work on the object. If the angle between the force and the object is $\frac{\pi}{3}$, what is the magnitude of the displacement of the object?
5. A force of $\left(2\hat{\imath} - 3\,\hat{\jmath} + 5\hat{k}\right)$ newton acting on an object gives a displacement of $\left(\hat{\imath} + 2\hat{\jmath} - \hat{k}\right)$ metres. How much work is done by the force?
6. In 3-dimensional space a force acts on an object and changes its position (in metre) from position A(3, 2, 6) to position B(7, 5, 15). If the force vector (in Newton) is $\underset{\sim}{\boldsymbol{F}} = 3\hat{\imath} - 2\,\hat{\jmath} + \hat{k}$, find the work done.
7. Vectors $\underset{\sim}{\boldsymbol{a}} = \hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\underset{\sim}{\mathbf{b}} = m\,\hat{\imath} - \hat{\jmath} + \hat{k}$ are at an angle of 60º to each other. Find the value of m.
8. If $\underset{\sim}{\mathbf{q}} = 3\,\hat{\imath} + 2\,\hat{\jmath}$ and $\underset{\sim}{\mathbf{r}} = \mathrm{c}\,\hat{\imath} - 2\,\hat{\jmath}$ find the value of c when $\underset{\sim}{\boldsymbol{q}}$ and $\underset{\sim}{\mathbf{r}}$ are parallel to each other (i.e. $\theta = 0$).
9. Solve each of the following equations:

   (a) $(\hat{\imath} + 2\hat{\jmath} - 3\hat{k}) \cdot (5\hat{\imath} + x\hat{\jmath} + \hat{k}) = 6$

   (b) $(x\hat{\imath} + 7\hat{\jmath} - \hat{k}) \cdot (-4\hat{\imath} + x\hat{\jmath} + 5\hat{k}) = 10$

   (c) $(x\hat{\imath} + 5\hat{k}) \cdot (-2\hat{\imath} - 3\hat{\jmath} + 3\hat{k}) = x$

(d) $x(2\hat{\imath} + 3\hat{\jmath} + \hat{k}) \cdot (\hat{\imath} + \hat{\jmath} + x\hat{k}) = 6$

10. $(x\hat{\imath} + 2\hat{\jmath} + y\hat{k})$ is perpendicular to vectors $(\hat{\imath} + \hat{\jmath} + \hat{k})$ and $(4\hat{\imath} + \hat{\jmath} + 2\hat{k})$. Find *x* and *y*.
11. Find the angle, in radians, between each of the following pairs of vectors, correct to 3 significant figures:

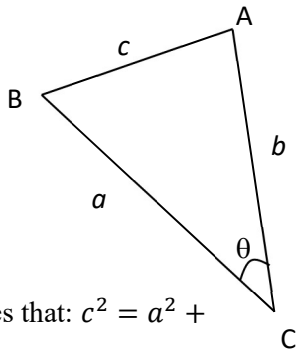
    (a) $(\hat{\imath} + 2\hat{\jmath} - \hat{k})$ and $(\hat{\imath} - 4\hat{\jmath} + \hat{k})$

    (b) $(-2\hat{\imath} + \hat{\jmath} + 3\hat{k})$ and $(-2\hat{\imath} - 2\hat{\jmath} + \hat{k})$

    (c) $(2\hat{\imath} - \hat{\jmath} - 3\hat{k})$ and $(4\hat{\imath} - 2\hat{k})$

    (d) $(7\hat{\imath} + \hat{k})$ and $(-\hat{\imath} + \hat{\jmath} - 3\hat{k})$

12. ABC is a triangle with dimensions as shown in the figure. $\angle ACB = \theta$.



    (a) Express $\overrightarrow{AB}$ in terms of $\overrightarrow{CA}$ and $\overrightarrow{CB}$.

    (b) Use the dot product $\overrightarrow{AB} \cdot \overrightarrow{AB}$ to prove the cosine rule which states that: $c^2 = a^2 + b^2 - 2ab\cos\theta$

13. (a) Find the component of the force $(7\hat{\imath} + 3\hat{\jmath})$ N in the direction of the vector $2\hat{\imath} - \hat{\jmath}$.

    (b) Find the component of the force $(2\hat{\imath} - 3\hat{\jmath})$ N in the direction of the vector $3\hat{\imath} + 4\hat{\jmath}$.

14. A floating ice block is pushed through a displacement $\vec{d} = (15m)\hat{\imath} - (12m)\hat{\jmath}$ along a straight embankment by rushing water, which exerts a f force $\vec{F} = (210N)\hat{\imath} - (150N)\hat{\jmath}$ on the block. How much work does the force do on the block during the displacement?
15. A coin slides over a frictionless plane and across an *xy* coordinate system from the origin to point with *xy* coordinates (3.0 m, 4.0 m) while a constant force acts on it. The force has magnitude of 2.0 N and is directed at a counterclockwise angle of 100° from the positive direction of the *x* axis. How much work is done by the force on the coin during the displacement? [Hint: You will want to sketch out the diagram.]

Exercise 5: Vector Operation: Cross Product

1. In each of the following find the cross product $\mathbf{p} \times \boldsymbol{q}$:

   (a) $\mathbf{p} = 3\,\hat{\imath} - \hat{\jmath}$; $\mathbf{q} = 2\,\hat{\imath} + \hat{k}$

   (b) $\mathbf{p} = 3\,\hat{\imath} - \hat{\jmath} + 2\,\hat{k}$; $\mathbf{q} = 2\,\hat{\imath} + 3\,\hat{\jmath} - \hat{k}$

   (c) $\mathbf{p} = \hat{\imath} + 3\hat{\jmath} - 4\hat{k}$; $\mathbf{q} = -2\,\hat{\imath} + \hat{\jmath} + 5\hat{k}$

   (d) $\mathbf{p} = 3\,\hat{\imath} - 4\,\hat{k}$; $\mathbf{q} = -4\,\hat{\imath} - 2\,\hat{\jmath} - 3\hat{k}$

2. Find the following cross products:

   (a) (1, 4, –6) × (3, 2, –4)

   (b) (–1, 0, 7) × (6, 4, 1)

3. Find the two unit vectors which are perpendicular to the plane containing the vectors :

   $\hat{\imath} - \hat{\jmath} + 2\,\hat{k}$ and $2\,\hat{\imath} + \hat{\jmath} - 3\hat{k}$.

4. A triangle ABC in 3-dimensional space has sides defined by the vectors. Use vector cross product to find the area of the triangle.

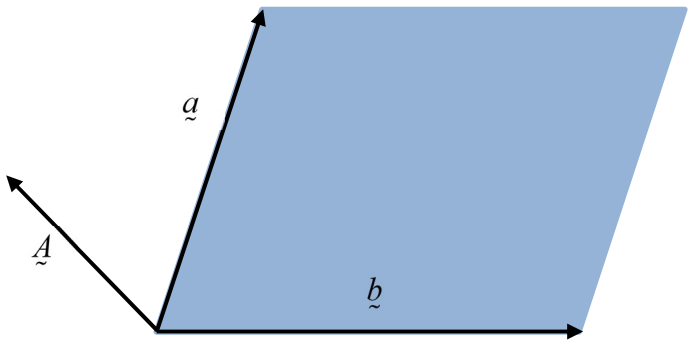
   $$\overrightarrow{\boldsymbol{AB}} = 2\hat{\imath} - 2\hat{\jmath} + \hat{k}\ ;\quad \overrightarrow{\boldsymbol{BC}} = -4\hat{\imath} + 2\hat{\jmath} - 2\hat{k}\ ;\ \overrightarrow{\boldsymbol{AC}} = -2\hat{\imath} - \hat{k}$$

5. A triangle in 3-dimensional space has vertices A(1, –1, 2), B(–1, 2, 1) and C(3, 1, 2).

   (a) Use a vector cross product to find the area of the triangle.

   (b) Find the unit vector perpendicular to the plane ABC

6. The area vector $\underset{\sim}{A}$ of a given surface is a vector which is pointing in the outward direction perpendicular to the surface and has a magnitude given by area of the surface. Given vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are vectors along the sides of a parallelogram (see diagram above), where $\underset{\sim}{a} = 3\hat{\imath} + 4\hat{\jmath} + \hat{k}$ and $\underset{\sim}{b} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}$. Find the area vector $\underset{\sim}{A}$ for the parallelogram and hence determine the area of the parallelogram. What is the angle between vector $\underset{\sim}{a}$ and $\underset{\sim}{b}$?



ANSWERS:

Exercise 1

1. $\overrightarrow{PT}$ 2. $\overrightarrow{AM}$ 3. 0 4. $\overrightarrow{AB}$

Exercise 2

1. (a) 47.2 m (b) 122°

2. 13.0 m , 7.5 m

3. (a) See lecturer

(b) $\overrightarrow{OP} = 4\hat{\imath}+2\hat{\jmath}$; $\overrightarrow{OQ} = -3\hat{\imath}+4\hat{\jmath}$; $\overrightarrow{OR} = -5\hat{\imath} - \hat{\jmath}$

(c) $\overrightarrow{QP}=7\hat{\imath} - 2\hat{\jmath}$; $\overrightarrow{RQ}= 2\hat{\imath} + 5\hat{\jmath}$; $\overrightarrow{PR}= -9\hat{\imath} - 3\hat{\jmath}$

(d) (i) $\sqrt{53}$ (ii) $\sqrt{29}$ (iii) $\sqrt{90}$

4. $7\hat{\imath} + 6\hat{\jmath}$; $-3\hat{\imath} + 2\hat{\jmath}$ **5.** $12\hat{\imath}$; $-2\hat{\imath} - 4\hat{\jmath}$ **6.** $2\hat{\imath} - 7\hat{\jmath}$

7. (a) $-\hat{\imath} - 7\hat{\jmath} + \hat{k}$ (b) $5\hat{k}$

8. (a) $x = 3, y = -\frac{1}{3}$ 8.(b) $x = 4, y = \frac{2}{5}$

9. (a) (i) $\overrightarrow{AB} = 4\hat{\imath} - 2\hat{\jmath} - 4\hat{k}$ 9. (a)(ii) $\overrightarrow{BC} = -5\hat{\imath} + 4\hat{\jmath} + 9\hat{k}$ 9.(a)(iii) $\overrightarrow{CD} = 2\hat{\imath} - \hat{\jmath} - 2\hat{k}$

9.(a) (iv) $\overrightarrow{DA} = -\hat{\imath} - \hat{\jmath} - 3\hat{k}$ 9. (b) (i) $\left|\overrightarrow{AC}\right| = \sqrt{30}$ 9.(b)(ii) $\left|\overrightarrow{BD}\right| = \sqrt{67}$

10.(a) (i)$\overrightarrow{OA} = 2\hat{\imath} + \hat{\jmath}$ 10. (a) (ii) $\overrightarrow{AB} = -\hat{\imath} - 4\hat{\jmath}$ 10.(a)(iii) $\overrightarrow{BC} = -6\hat{\imath} + 5\hat{\jmath}$

10.(a) (iv) $\overrightarrow{BD} = 2\hat{\imath} + 8\hat{\jmath}$

10. (b) If $\overrightarrow{AB}$ and $\overrightarrow{BD}$are parallel, $\overrightarrow{BD} = n\overrightarrow{AB}$,where n is a real number

10. (c) A, B and D are collinear (i.e. points on the same straight line)

11.$(11\hat{\imath} - 3\hat{\jmath})$ N

12. (a) (i) $(1.38\hat{\imath} + 5.39\hat{\jmath})$ N , (ii) 5.56 N, 75.6°

12. (b) (i) $(2.19\hat{\imath} - 2.19\hat{\jmath})$ N , (ii) 3.10 N, -45.0° or 315°

12. (c) (i) $(-2.15\hat{\imath} - 1.01\hat{\jmath})$ N , (ii) 2.38 N, -154.8° or 205.2°

Exercise 3

1. (a) $\sqrt{41}$ (b) $\frac{1}{\sqrt{41}}(4\,\hat{\imath}\text{ - }5\,\hat{\jmath})$ (c) -51.3° (d) $-6\hat{\imath} + 8\hat{\jmath}$

2. (a) $\underset{\sim}{\mathbf{p}} = 3\hat{\imath} - \hat{\jmath} + 4\hat{k}$; $\underset{\sim}{\mathbf{q}} = 2\hat{\imath} + 3\hat{\jmath} + \hat{k}$; $\underset{\sim}{\mathbf{r}} = \hat{\imath} - 3\hat{\jmath} - 2\hat{k}$ (b) $\frac{1}{\sqrt{14}}(2\hat{\imath} + 3\hat{\jmath} + \hat{k})$

(c) $5\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$ (d) $-\hat{\imath} - 5\hat{\jmath} - 8\hat{k}$

(e) $\cos\alpha = \frac{1}{\sqrt{14}}$; $\cos\beta = \frac{-3}{\sqrt{14}}$; $\cos\gamma = \frac{-2}{\sqrt{14}}$ (f) $\alpha = 54.0°;\ \beta = 101.3°;\ \gamma = 38.3°$

(g) $-\hat{\imath} + 4\hat{\jmath} - 3\hat{k}$ (h) $2\hat{\imath} + 2\hat{\jmath} + 6\hat{k}$ (i) 14.4

3. $\frac{1}{\sqrt{30}}(5\,\hat{\imath} + 2\,\hat{\jmath} - \hat{k})$4. $m = \frac{1}{6}$, $n = -\frac{11}{12}$

5.(a) (i) $|\underset{\sim}{a}| = \sqrt{29}$ 5.(a) (ii) $|\underset{\sim}{b}| = \sqrt{13}$ 5.(a)(iii) $|\underset{\sim}{a} + 2\underset{\sim}{b}| = \sqrt{97}$ 5.(a)(iv) $|\underset{\sim}{c} - \underset{\sim}{d}| = \sqrt{19}$

5.(b)(i) 21.80° 5.(b) (ii) 23.96° 5.(b)(iii) 46.51°

Exercise 4

1. (a) -13 (b) -13 (c) $121.2°$ (d) Show that $\underset{\sim}{\mathbf{p}} \cdot \underset{\sim}{\mathbf{r}} = 0$; so $cos\,\theta = 0$

(e) $\sqrt{14}$ (f) 14 (g) 13

2. $c = \frac{4}{3}$ **3**. 122.9° **4**. 22 metre **5**. –9 joule

6. 15 joule **7**. $m = 2$ **8**. $c = -3$

9. (a) $x = 2$ 9. (b) $x = 5$ 9. (c) $x = 5$ 9. (d) $x = -6, x = 1$

10. $x = 1, y = -3$

11. (a) 2.45 11. (b) 1.11 11. (c) 0.580 11. (d) 2.01

12. See lecturer for explanation

13 (a) $\frac{11}{5}(2\hat{\imath} - \hat{\jmath})$ 13. (b) $-\frac{6}{25}(3\hat{\imath} + 4\hat{\imath})$

14. 4950 J

15. 6.8 J.

Exercise 5

1. $(a)\ -\hat{\imath} - 3\hat{\jmath} + 2\hat{k}\ \ (b)\ -5\hat{\imath} + 7\hat{\jmath} + 11\hat{k}\ \ (c)\ 19\hat{\imath} + 3\hat{\jmath} + 7\hat{k}\ \ (d)\ -8\hat{\imath} + 25\hat{\jmath} - 6\hat{k}$

2. $(a)\ -2(2\hat{\imath} + 7\hat{\jmath} + 5\hat{k})\ \ (b)\ -28\hat{\imath} + 43\hat{\jmath} - 4\hat{k}$

3. $\frac{1}{\sqrt{59}}(\hat{\imath} + 7\hat{\jmath} + 3\hat{k})$, $-\frac{1}{\sqrt{59}}(\hat{\imath} + 7\hat{\jmath} + 3\hat{k})$ **4**. $\sqrt{5}$ square units

5. (a) $3\sqrt{3}$ square units (b) $\frac{1}{3\sqrt{3}}(\hat{\imath} - \hat{\jmath} - 5\hat{k})$ or $-\frac{1}{3\sqrt{3}}(\hat{\imath} - \hat{\jmath} - 5\hat{k})$

6. $7\hat{\imath} - 5\hat{\jmath} - \hat{k}$, $5\sqrt{3}$ units^2 , 27.0^o